TMUA / WORKED SOLUTIONS

Set A Paper 2

Mathematical Reasoning 20 questions

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Proof & Error Analysis

1. Let x and y be real numbers.

A student attempts to prove the following claim:

$$\textbf{Claim: If } x + y \geq 2 \text{ then } x^2 + y^2 \geq 2$$

The student's attempt is as follows:

Line 1: Since $x + y \geq 2$, at least one of x and y is at least 1.
Assume $x \geq 1$; the other case is the same with x and y swapped.

Line 2: From $x + y \geq 2$ we get $y \geq 2 - x$

Line 3: Therefore $y^2 \geq (2 - x)^2$

Line 4: Hence $x^2 + y^2 \geq x^2 + (2 - x)^2$

Line 5: Now $x^2 + (2 - x)^2 = 2(x - 1)^2 + 2$, and this is at least 2.

Line 6: Therefore $x^2 + y^2 \geq 2$, as claimed.

Which of the following best describes the argument?

A Line 1 contains the first error.

B Line 2 contains the first error.

C Line 3 contains the first error.

D Line 4 contains the first error.

E Line 5 contains the first error.

F The argument is a correct proof of the claim.

G The argument contains no error, but the claim is false.

Solution **Answer C**

The claim looks plausible. No counterexample is readily apparent. Instead, let's check each line in order.

Line 1 is valid: if both $x < 1$ and $y < 1$, then $x + y < 2$. Swapping x and y does not change the claim, so it is enough to consider $x \geq 1$.

Line 2 is also valid: subtracting x from $x + y \geq 2$ gives $y \geq 2 - x$.

Line 3 is the first error. Squaring does not always preserve an inequality when negative numbers are involved.

For example, take $x = 3$ and $y = -\frac{1}{2}$. Then $2 - x = -1$, $(2 - x)^2 = 1$ and $y^2 = \frac{1}{4}$.

Now we see that squaring does **not** maintain the direction of the inequality:

$$-\frac{1}{2} \geq -1$$

but

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4} < (-1)^2 = 1$$

So squaring reversed the inequality. This is a counterexample to the step on line 3, not to the claim itself.

Therefore the first error is on line 3, so the correct option is **C**.

Remark: (Optional)

To see why the claim is true despite the faulty proof, note the algebraic fact:

$$\begin{aligned}(x+y)^2 + (x-y)^2 &= (x^2 + 2xy + y^2) + (x^2 - 2xy + y^2) \\ &= 2x^2 + 2y^2 \\ &= 2(x^2 + y^2)\end{aligned}$$

so $(x+y)^2 + (x-y)^2 = 2(x^2+y^2)$

Since $(x-y)^2 \geq 0$ it follows that

$$2(x^2+y^2) = (x+y)^2 + (x-y)^2 \geq (x+y)^2$$

So $2(x^2+y^2) \geq (x+y)^2$ or equivalently $x^2 + y^2 \geq \dfrac{(x+y)^2}{2}$

Finally, we are told that $x + y \geq 2$ so $(x+y)^2 \geq 2^2 = 4$ (we are allowed to square an inequality when both sides are positive)

Therefore

$$x^2 + y^2 \geq \frac{2^2}{2} = 2$$

Thus, it is true that if $x + y \geq 2$ then $x^2 + y^2 \geq 2$, but the given argument is not a correct proof of that fact.

Providing a correct proof was, of course, not a requirement of the question. But it explains why counterexamples were hard to find. False proofs of correct statements are some of the hardest error analysis questions.

Trigonometry

2. In quadrilateral $ABCD$, the diagonal AC divides the figure into two right-angled triangles, $\triangle ABC$ and $\triangle ACD$.

- In $\triangle ABC$, $\angle ABC = 90°$, $\angle BAC = 30°$, and $AB = \sqrt{6}$ cm
- In $\triangle ACD$, $\angle ACD = 90°$, and $\tan(\angle CAD) = \dfrac{1}{\sqrt{2}}$

What is the area of quadrilateral $ABCD$, in cm^2?

A $\sqrt{3} + \sqrt{2}$

B $\sqrt{3} + 2\sqrt{2}$

C $2\sqrt{3} + \sqrt{2}$

D $2\sqrt{3} + 2\sqrt{2}$

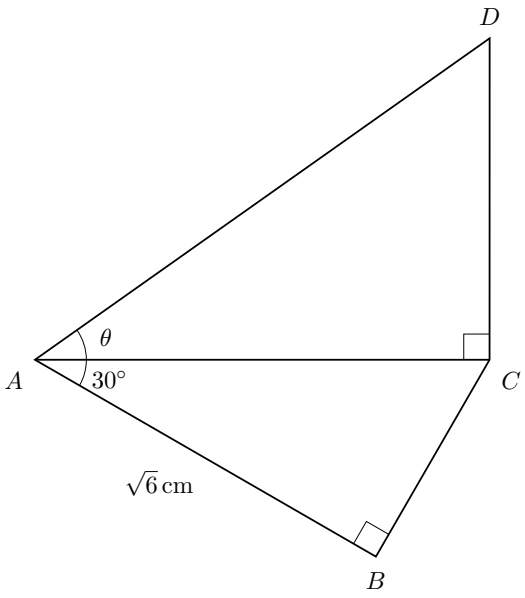
E $\sqrt{3} + 4\sqrt{2}$

F $2\sqrt{3} + 4\sqrt{2}$

Solution **Answer B**

Let $\theta = \angle CAD$, so $\tan\theta = \dfrac{1}{\sqrt{2}}$

The situation can be explained by the diagram:



In right-angled triangle ABC, the side opposite the 30° angle is BC and the adjacent side is AB. Hence

$$\tan 30° = \frac{BC}{AB} = \frac{BC}{\sqrt{6}} = \frac{1}{\sqrt{3}}$$

Therefore

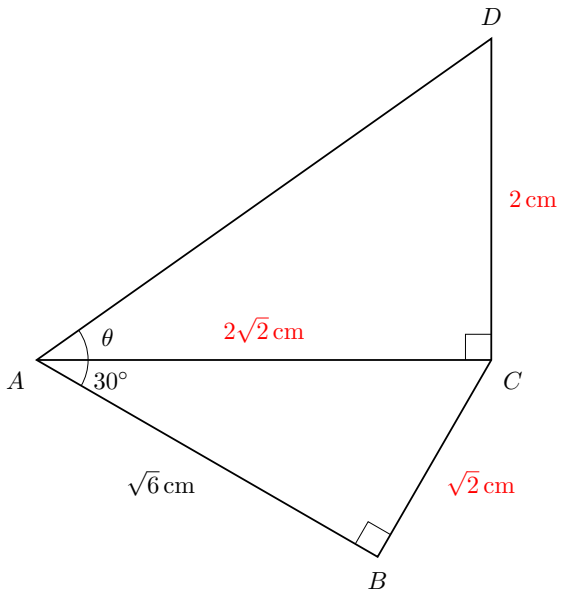
$$BC = \frac{\sqrt{6}}{\sqrt{3}} = \sqrt{2} \text{ cm}$$

Pythagoras now gives the shared diagonal:

$$AC^2 = AB^2 + BC^2 = 6 + 2 = 8 \quad \Longrightarrow \quad AC = 2\sqrt{2} \text{ cm}$$

In right-angled triangle ACD, CD is opposite θ and AC is adjacent to it. Thus

$$\tan\theta = \frac{CD}{AC} = \frac{CD}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \quad \Longrightarrow \quad CD = 2 \text{ cm}$$



The two right-angled triangles have perpendicular side pairs AB, BC and AC, CD. Their areas are

$$\text{Area of } ABC = \frac{1}{2}AB \times BC = \frac{1}{2}\sqrt{6} \times \sqrt{2} = \sqrt{3} \text{ cm}^2,$$
$$\text{Area of } ACD = \frac{1}{2}AC \times CD = \frac{1}{2}(2\sqrt{2}) \times 2 = 2\sqrt{2} \text{ cm}^2.$$

Adding them gives $\sqrt{3} + 2\sqrt{2}$ cm^2, so the correct option is **B**.

Proof & Error Analysis

3. Consider the following claim and attempted proof.

Claim: Let $f(x)$ be a quadratic polynomial. If $\int_{-1}^{1} f(x)\,\mathrm{d}x = 0$, then the graph of $y = f(x)$ must intersect the x-axis at two distinct points in the interval $-1 < x < 1$.

Attempted proof:

Line 1 Let $f(x) = ax^2 + bx + c$, where a, b, c are real constants with $a \neq 0$

Line 2 Evaluating the integral gives $\int_{-1}^{1} f(x)\,\mathrm{d}x = \frac{2a}{3} + 2c = 0$, so $c = -\frac{a}{3}$

Line 3 The discriminant is $b^2 - 4ac = b^2 + \frac{4a^2}{3} > 0$, so $f(x) = 0$ has two distinct real roots

Line 4 Writing these roots as x_1 and x_2, we have $x_1 x_2 = \frac{c}{a} = -\frac{1}{3}$

Line 5 Since $|x_1 x_2| = \frac{1}{3} < 1$, it follows that $|x_1| < 1$ and $|x_2| < 1$

Line 6 Therefore, both roots lie in the interval $-1 < x < 1$

Which line of the proof contains the first error?

A Line 1

B Line 2

C Line 3

D Line 4

E Line 5

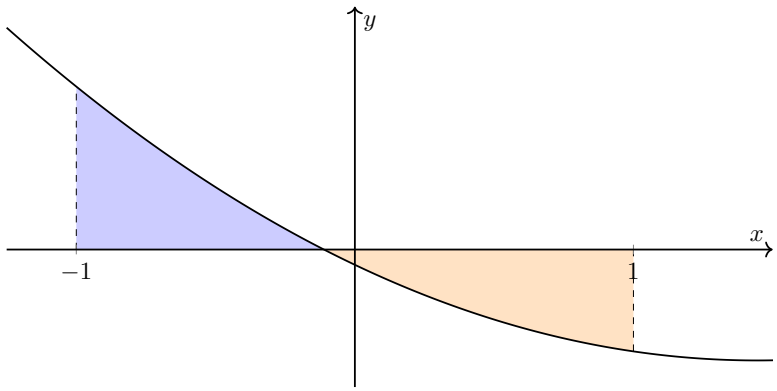
F Line 6

G There is no error, the proof is correct as written.

Solution **Answer E**

An integral can be zero when the area above the x-axis balances the area below it.

The sketch shows how this might happen for a quadratic with only one root between -1 and 1, indicating that the claim is false:



We could search for a specific quadratic with this property and plug it in line-by-line to find the error, but finding the quadratic itself sounds like a lot of effort.

Therefore, we will analyse the proof line-by-line.

Line 1 gives the general form of a quadratic. $a \neq 0$ is required so the x^2 term doesn't cancel. Line 1 is correct.

In Line 2, the bx term contributes zero because its integral from -1 to 1 cancels on either side of 0. Thus

$$\int_{-1}^{1} (ax^2 + bx + c)\,\mathrm{d}x = \left[\frac{ax^3}{3} + \frac{bx^2}{2} + cx\right]_{-1}^{1} = \frac{2a}{3} + 2c = 0$$

so $c = -a/3$, as stated. Line 2 is correct.

Using this value of c, Line 3 gives

$$b^2 - 4ac = b^2 + \frac{4a^2}{3} > 0$$

because $a \neq 0$. There are therefore two distinct real roots because the discriminant is strictly positive. Line 3 is correct.

Line 4 correctly uses the product of the roots formula, $x_1 x_2 = \dfrac{c}{a}$ to obtain

$$x_1 x_2 = -\frac{1}{3}$$

The mistake is in Line 5: a product with absolute value less than 1 does not require both factors to have absolute value less than 1.

For example, $x_1 = 3$ and $x_2 = \dfrac{1}{9}$ give $|x_1 x_2| = \dfrac{1}{3}$, although $|x_1| = 3 > 1$

Even retaining the negative product from Line 4, $x_1 = 3$ and $x_2 = -\dfrac{1}{9}$ give $x_1 x_2 = -\dfrac{1}{3}$, with one root outside the interval $-1 \leq x \leq 1$

The first error is therefore in Line 5, so the correct option is **E**.

Exponentials & Logarithms

4. Let a be a positive real number with $a \neq 1$

The equation

$$\log_a(3x-1) = 2 + \log_a(x+1)$$

has exactly one real solution x

What is the complete range of all the possible values of a?

A $0 < a < \sqrt{3}$ and $a \neq 1$

B $a > 1$

C $a > 0$ and $a \neq 1$

D No such a exists

E $a = 3$

F $a = \frac{1}{3}$

G $a > 0$, $a \neq 1$ and $a \neq \sqrt{3}$

Solution **Answer A**

Subtracting $\log_a(x+1)$ from both sides gives

$$\log_a(3x-1) - \log_a(x+1) = 2$$

Combining the logarithms,

$$\log_a\left(\frac{3x-1}{x+1}\right) = 2$$

Hence, provided the original logarithms are defined,

$$\frac{3x-1}{x+1} = a^2$$

Multiplying through and rearranging,

$$3x - 1 = a^2x + a^2$$

$$(3-a^2)x = 1 + a^2$$

Hence, provided $3 - a^2 \neq 0$, there is only one candidate solution

$$x = \frac{1+a^2}{3-a^2}$$

If $a = \sqrt{3}$, then $3 - a^2 = 0$ and the equation $(3-a^2)x = 1 + a^2$ becomes $0 = 4$, which has no solution, so $a = \sqrt{3}$ must be excluded from the answer.

Excluding $a = \sqrt{3}$ does not yet give option **G**: our candidate solution must also make both original logarithms defined (that is, we must make sure the insides of both logarithms are positive).

In particular, we must check this before deciding whether the range is **A** or a smaller set such as **F**.

Substituting the candidate solution into each logarithm's argument gives

$$x+1=\frac{4}{3-a^2} \qquad \text{and} \qquad 3x-1=\frac{4a^2}{3-a^2}$$

Both must be positive to ensure the original logarithms are defined.

Because $a>0$, both numerators are positive, so each condition requires $3-a^2>0$. This gives $a<\sqrt{3}$.

Conversely, every a with $0<a<\sqrt{3}$ and $a\neq 1$ makes both arguments positive and gives the unique solution found above.

Hence the complete range is

$$0<a<\sqrt{3}, \qquad a\neq 1$$

The correct option is **A**.

Number

5. A positive integer n can be written as $n = 2^a 3^b$ where a and b are non-negative integers.

The number n has exactly 12 positive factors, counting 1 and n itself.

Given that $n < 200$, what is the greatest possible value of n?

A 72

B 96

C 108

D 144

E 162

F 486

Solution **Answer C**

Since

$$n = 2^a 3^b$$

every positive factor of n is obtained by choosing a power of 2 from

$$2^0, 2^1, \ldots, 2^a$$

and independently choosing a power of 3 from

$$3^0, 3^1, \ldots, 3^b$$

Thus every positive factor has the form $2^i 3^j$, where there are $a+1$ possible values of i and $b+1$ possible values of j.

Therefore the total number of positive factors is $(a+1)(b+1)$ and the question then gives us that

$$(a+1)(b+1) = 12$$

Test the choices from largest to smallest: the first one below 200 that satisfies this equation must be the answer.

Choice F is $486 > 200$, so it is too large.

For the next three choices,

$$\begin{aligned} 162 = 2^1 3^4 &\quad \text{gives } (1+1)(4+1) = 10 \neq 12 \\ 144 = 2^4 3^2 &\quad \text{gives } (4+1)(2+1) = 15 \neq 12 \\ 108 = 2^2 3^3 &\quad \text{gives } (2+1)(3+1) = 12 \checkmark \end{aligned}$$

Thus 108 is the greatest eligible choice, so the answer is **C**.

Alternative Solution:

Let's try to obtain 108 without testing the given options.

Since $2^8 = 256$ and $3^5 = 243$, the condition $n < 200$ requires $a \leq 7$ and $b \leq 4$.

The possible factor pairs for $(a+1)(b+1) = 12$ are 1 and 12, 2 and 6, or 3 and 4, in either order.

The bounds $a \leq 7$ and $b \leq 4$ rule out all but these cases:

a	b	$2^a 3^b$
2	3	108
3	2	72
5	1	96

The largest is 108, so the answer is again **C**.

Testing the solutions from the largest first is clearly more efficient here. Even if it feels a little like cheating, it is an entirely valid and highly strategic move in a time-pressured multiple choice test.

Proof & Error Analysis

6. Let a and b be real numbers. A student attempts to prove the following claim:

Claim: If $a+b$ and ab are both integers, then a and b are both integers.

The student's attempt is as follows:

I a and b are the two solutions of $t^2-(a+b)t+ab=0$

II Write $s=a+b$ and $p=ab$, so s and p are integers and the two solutions are $t=\dfrac{s\pm\sqrt{s^2-4p}}{2}$

III Since a and b are real, $s^2-4p\geq 0$

IV Also s^2-4p is an integer, so $\sqrt{s^2-4p}$ is either an integer or an irrational number.

V If $\sqrt{s^2-4p}$ were irrational, then a would be irrational, and so $a+b$ would be irrational, contradicting the fact that $a+b$ is an integer.

VI Hence $\sqrt{s^2-4p}$ is an integer, so $s\pm\sqrt{s^2-4p}$ is an integer, and therefore a and b are both integers.

Which of the following best describes this attempt?

A It is completely correct.

B It is incorrect, and the first error is on line I.

C It is incorrect, and the first error is on line II.

D It is incorrect, and the first error is on line III.

E It is incorrect, and the first error is on line IV.

F It is incorrect, and the first error is on line V.

G It is incorrect, and the first error is on line VI.

Solution **Answer F**

A counterexample to the claim is $a=\sqrt{2}$ and $b=-\sqrt{2}$.

Neither is an integer, but

$$a+b=0, \qquad ab=-2$$

Both required quantities are integers, so we can run this example through the student's argument to find the first faulty line.

Line I works. With these choices the quadratic becomes t^2-2 which clearly has roots $\pm\sqrt{2}$:

$$\begin{aligned} t^2-(a+b)t+ab &= t^2-2 \\ &= (t-\sqrt{2})(t+\sqrt{2}) \end{aligned}$$

Line II also works. The quadratic formula gives the same result for the roots:

$$t = \frac{0 \pm \sqrt{0^2 - 4(-2)}}{2} = \pm\sqrt{2}$$

Lines III and IV also hold. The discriminant is

$$s^2 - 4p = 0^2 - 4(-2) = 8 \geq 0$$

The square root of the discriminant $\sqrt{s^2 - 4p} = \sqrt{8} = 2\sqrt{2}$ is irrational, as line IV allows.

The first error is on line V:

Although $a = \sqrt{2}$ is irrational, it does not follow that $a + b$ is irrational: b is irrational too, and the two terms cancel to give $a + b = 0$, a rational number.

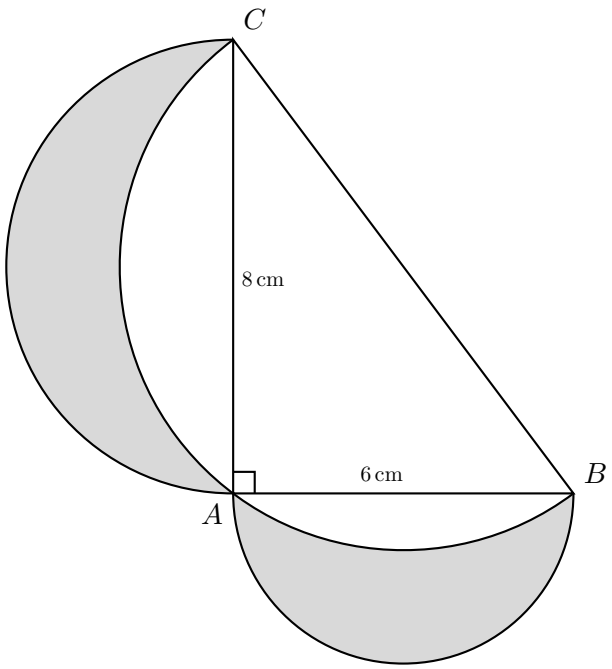
The student's argument first draws a false conclusion from true statements on line V, so the correct option is **F**.

Geometry

7. Triangle ABC is right-angled at A, with $AB = 6\,\text{cm}$ and $AC = 8\,\text{cm}$.

Semicircles are drawn with AB, AC and BC as diameters, as shown. The semicircle on BC is on the same side of the line BC as A. The portions of the other two semicircles that lie outside the semicircle on BC are shaded.

What is the total shaded area, in cm^2?



A 12

B 24

C 30

D 40

E 48

F 50

Solution **Answer B**

By Pythagoras,

$$BC = \sqrt{6^2 + 8^2} = 10$$

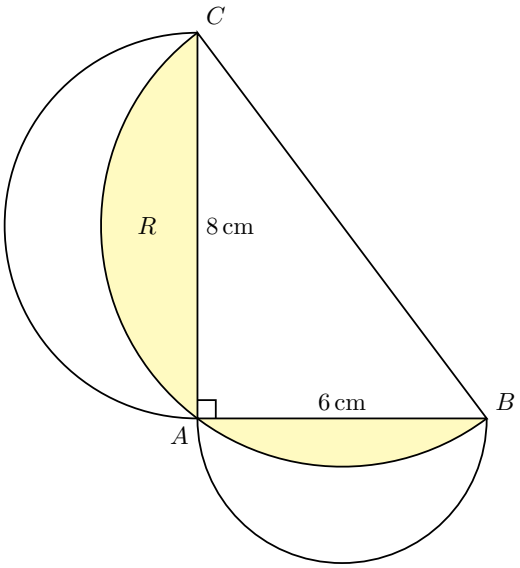
The semicircle on BC therefore has radius 5, so its area is

$$\frac{1}{2}\pi(5^2) = \frac{25\pi}{2}$$

The area of triangle ABC is

$$\frac{1}{2} \times 6 \times 8 = 24$$

Let R be the part of the semicircle on BC that lies outside triangle ABC, comprising of two regions shaded yellow in the figure below:



The area of R is therefore

$$R = \frac{25\pi}{2} - 24$$

The two smaller semicircles have radii 3 and 4, so their combined area is

$$\frac{1}{2}\pi(3^2) + \frac{1}{2}\pi(4^2) = \frac{25\pi}{2}$$

The required shaded regions are precisely the parts of these two smaller semicircles that remain after removing R. Hence their total area is

$$\frac{25\pi}{2} - R = \frac{25\pi}{2} - \left(\frac{25\pi}{2} - 24\right) = 24$$

Thus the total shaded area is $24\,\text{cm}^2$, exactly the same as the area of triangle ABC.

The correct option is **B**.

Remark: (optional)

This geometry is not special to the 6-8-10 triangle.

For any right-angled triangle, the area of a semicircle is proportional to the square of its diameter, so Pythagoras gives:

area of the two semicircles on the shorter sides = area of the semicircle on the hypotenuse

Subtracting the parts common to these semicircles leaves the two curved regions outside the hypotenuse semicircle on one side, and the original right-angled triangle on the other. Hence the two curved regions always have total area equal to the area of the triangle.

These regions are known as the *lunes of Hippocrates*. The result is a classical example of a curved shape whose area can be found exactly by reducing it to the area of a rectilinear figure.

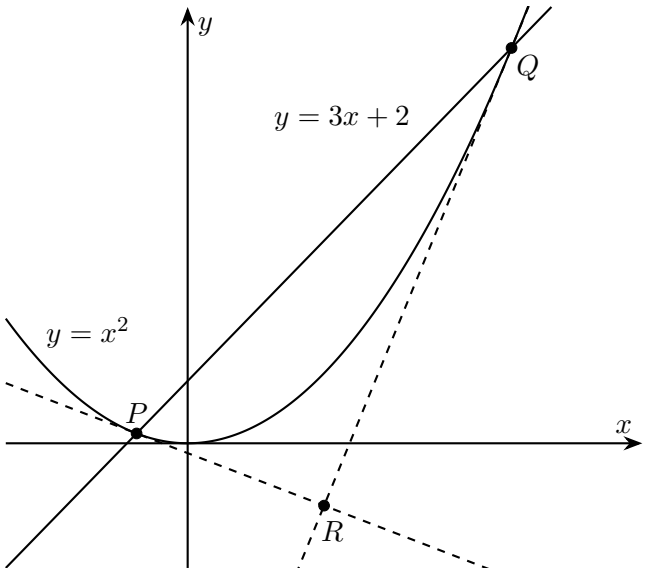
It is also closely connected to the Pythagorean theorem: the familiar relation

$$a^2 + b^2 = c^2$$

can equally well be expressed using the areas of any three similar figures constructed on the sides of a right-angled triangle, including semicircles.

Coordinate Geometry

8. The line $y = 3x + 2$ meets the parabola $y = x^2$ at the points P and Q.
The tangents to the parabola at P and Q meet at R, as shown.
What are the coordinates of R?



A $\left(\frac{3}{2}, -4\right)$
B $\left(\frac{3}{2}, -2\right)$
C $\left(\frac{3}{2}, -1\right)$
D $(3, -2)$
E $(1, -3)$
F $\left(2, -\frac{3}{2}\right)$

Solution **Answer B**

Let the x-coordinates of P and Q be p and q respectively.
Since P and Q lie on both $y = x^2$ and $y = 3x + 2$, their x-coordinates satisfy

$$x^2 = 3x + 2$$

so

$$x^2 - 3x - 2 = 0$$

Recall that the sum of the roots of a quadratic is $-\frac{b}{a}$ and the product of the roots is $\frac{c}{a}$
Hence, for $x^2 - 3x - 2 = 0$,

$$p + q = -\frac{-3}{1} = 3$$

and

$$pq = \frac{-2}{1} = -2$$

Since $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$ the tangent to $y = x^2$ at the point (p, p^2) has gradient $2p$, so its equation is

$$y - p^2 = 2p(x - p)$$

or

$$y = 2px - p^2$$

Similarly, the tangent at (q, q^2) is

$$y = 2qx - q^2$$

At their intersection R, these two expressions for y are equal, so

$$2px - p^2 = 2qx - q^2$$

Rearranging,

$$2x(p - q) = p^2 - q^2 = (p - q)(p + q)$$

Since P and Q are distinct, $p \neq q$, so

$$x = \frac{p + q}{2}$$

Thus the x-coordinate of R is the average of the two roots. Since the sum of the roots is

$$p + q = -\frac{b}{a} = 3$$

we have

$$x = \frac{3}{2}$$

Substituting $x = \frac{p + q}{2}$ into the tangent at P gives

$$y = 2p\left(\frac{p + q}{2}\right) - p^2$$

so

$$y = p(p + q) - p^2 = pq$$

Thus the y-coordinate of R is the product of the two roots. Since

$$pq = \frac{c}{a} = -2$$

we have

$$y = -2$$

Hence

$$R = \left(\frac{3}{2}, -2\right)$$

The correct option is **B**.

Alternative Solution: (Brute force)

Instead of using the sum and product of the roots, we can find the points P and Q explicitly and then calculate the intersection of their tangents. We will see that this method proves to be quite painful.

At P and Q,

$$x^2 = 3x + 2$$

so

$$x^2 - 3x - 2 = 0$$

Using the quadratic formula,

$$x = \frac{3 \pm \sqrt{17}}{2}$$

Hence

$$P = \left(\frac{3 - \sqrt{17}}{2}, \left(\frac{3 - \sqrt{17}}{2} \right)^2 \right)$$

and

$$Q = \left(\frac{3 + \sqrt{17}}{2}, \left(\frac{3 + \sqrt{17}}{2} \right)^2 \right)$$

Squaring the x-coordinates gives

$$P = \left(\frac{3 - \sqrt{17}}{2}, \frac{13 - 3\sqrt{17}}{2} \right)$$

and

$$Q = \left(\frac{3 + \sqrt{17}}{2}, \frac{13 + 3\sqrt{17}}{2} \right)$$

Since

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$$

the gradient of the tangent at P is

$$3 - \sqrt{17}$$

and the gradient of the tangent at Q is

$$3 + \sqrt{17}$$

Using the tangent equation $y = 2tx - t^2$ at a point (t, t^2), the tangent at P is

$$y = (3 - \sqrt{17})x - \frac{13 - 3\sqrt{17}}{2}$$

while the tangent at Q is

$$y = (3 + \sqrt{17})x - \frac{13 + 3\sqrt{17}}{2}$$

At their intersection R, the two values of y are equal, so

$$(3 - \sqrt{17})x - \frac{13 - 3\sqrt{17}}{2} = (3 + \sqrt{17})x - \frac{13 + 3\sqrt{17}}{2}$$

Rearranging gives

$$3\sqrt{17} = 2\sqrt{17}\,x$$

and therefore

$$x = \frac{3}{2}$$

Substituting into either tangent equation,

$$y = (3 - \sqrt{17})\frac{3}{2} - \frac{13 - 3\sqrt{17}}{2}$$

so

$$y = \frac{9 - 3\sqrt{17} - 13 + 3\sqrt{17}}{2} = -2$$

Hence

$$R = \left(\frac{3}{2}, -2\right)$$

So we again obtain option **B** but with considerably more pain and suffering.

Remark: (Optional)

There is a general property of parabolas behind the calculation above.

If two points on a general parabola with equation $y = Ax^2 + Bx + C$ have x-coordinates p and q, then the tangents at those points meet at a point whose x-coordinate is

$$\frac{p+q}{2}$$

This is the same as the x-coordinate of the midpoint of the chord joining the two points.

Geometrically, the intersection of the two tangents and the midpoint of the chord therefore lie on a line parallel to the axis of the parabola.

Logic

9. Consider the statement $(*)$:

$$\textbf{If } 0 \leq f(x) \leq 1 \text{ for all real } x \text{ with } 0 \leq x \leq 1, \textbf{ then}$$
$$\int_0^1 f(x)\,\mathrm{d}x \leq \int_0^1 (f(x))^2\,\mathrm{d}x \qquad (*)$$

Which of the following functions is a **counterexample** to $(*)$?

A $f(x) = 2x$

B $f(x) = x^2 + \frac{1}{2}$

C $f(x) = 1$

D $f(x) = 1 + x$

E $f(x) = 0$

F $f(x) = 5x(1-x)$

G $f(x) = \sqrt{x} + \frac{1}{2}$

H $f(x) = \frac{1}{2}$

Solution **Answer H**

A counterexample to a statement of the form "if A, then B" must make the antecedent A true and the consequent B false.

Here, the antecedent is the condition

$$0 \leq f(x) \leq 1$$

for every x with $0 \leq x \leq 1$

The consequent is the inequality

$$\int_0^1 f(x)\,\mathrm{d}x \leq \int_0^1 (f(x))^2\,\mathrm{d}x$$

So a function that fails the inequality $0 \leq f(x) \leq 1$ cannot be a counterexample, regardless of what happens to its integrals. It is therefore quickest to check the antecedent first.

At $x = 1$, four options immediately fail:

$$\text{A:} \quad f(1) = 2(1) = 2 > 1$$
$$\text{B:} \quad f(1) = 1^2 + \frac{1}{2} = \frac{3}{2} > 1$$
$$\text{D:} \quad f(1) = 1 + 1 = 2 > 1$$
$$\text{G:} \quad f(1) = \sqrt{1} + \frac{1}{2} = \frac{3}{2} > 1$$

Option F also fails the antecedent inequality. Taking $x = \frac{1}{2}$,

$$f\left(\frac{1}{2}\right) = 5\left(\frac{1}{2}\right)\left(1 - \frac{1}{2}\right) = \frac{5}{4} > 1$$

Thus A, B, D, F and G all make the antecedent false, so none of them can be a counterexample.

This leaves C, E and H. These are the constant functions

$$f(x) = 1, \qquad f(x) = 0, \qquad f(x) = \frac{1}{2}$$

and all three satisfy $0 \leq f(x) \leq 1$ throughout $0 \leq x \leq 1$

For C,

$$f(x) = 1 \quad \Longrightarrow \quad (f(x))^2 = 1$$

so

$$\int_0^1 f(x)\,\mathrm{d}x = \int_0^1 (f(x))^2\,\mathrm{d}x = 1$$

The required inequality is true, so C is not a counterexample.

Similarly, for E,

$$f(x) = 0 \quad \Longrightarrow \quad (f(x))^2 = 0$$

and therefore

$$\int_0^1 f(x)\,\mathrm{d}x = \int_0^1 (f(x))^2\,\mathrm{d}x = 0$$

Again, the required inequality is true, so E is not a counterexample.

For H, $f(x) = \frac{1}{2}$, so the antecedent is certainly true. However,

$$\int_0^1 \frac{1}{2}\,\mathrm{d}x = \frac{1}{2}$$

whereas

$$\int_0^1 \left(\frac{1}{2}\right)^2 \mathrm{d}x = \int_0^1 \frac{1}{4}\,\mathrm{d}x = \frac{1}{4}$$

The claimed conclusion would therefore require

$$\frac{1}{2} \leq \frac{1}{4}$$

which is false.

Thus **H** makes the antecedent true and the consequent false, so it is a counterexample to the statement.

Logic

10. Consider the following three conditions on a positive real number x

$$\begin{aligned}&(1)\quad x^2-3x+2<0\\&(2)\quad \log_{10}x>0\\&(3)\quad 2^x>4\end{aligned}$$

Which of the following statements is/are true?

I: Condition (2) holds only if at least one of conditions (1) and (3) holds

II: If condition (2) does not hold, then condition (3) does not hold

III: For condition (1) to hold, it is necessary but not sufficient that condition (3) does not hold

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Solution **Answer G**

First translate each of the three conditions into a simple condition on x.

For condition (1),

$$x^2-3x+2<0$$

so

$$(x-1)(x-2)<0$$

The product is negative between the two roots, hence

$$(1)\qquad 1<x<2$$

For condition (2), since $\log_{10}x$ is increasing and $\log_{10}1=0$,

$$\log_{10}x>0$$

is equivalent to

$$(2)\qquad x>1$$

For condition (3), since $4 = 2^2$ and 2^x is increasing,

$$2^x > 4$$

is equivalent to

$$(3) \qquad x > 2$$

So the three conditions are simply

$$(1) \quad 1 < x < 2$$
$$(2) \quad x > 1$$
$$(3) \quad x > 2$$

We can now translate each statement into ordinary inequalities.

For statement I, the phrase "condition (2) holds only if at least one of conditions (1) and (3) holds" means

$$\text{if (2) holds, then (1) or (3) must hold}$$

In terms of x, this says

$$x > 1 \quad \Longrightarrow \quad 1 < x < 2 \text{ or } x > 2$$

This is almost true, but $x = 2$ is missed. At $x = 2$, condition (2) holds because $2 > 1$, but condition (1) does not hold because $2 \not< 2$, and condition (3) does not hold because $2 \not> 2$.

Therefore statement I is false.

For statement II,

$$\text{if condition (2) does not hold, then condition (3) does not hold}$$

Since x is positive, condition (2) not holding means

$$0 < x \leq 1$$

Condition (3) not holding means

$$x \leq 2$$

So statement II is asking whether

$$0 < x \leq 1 \quad \Longrightarrow \quad x \leq 2$$

This is always true. Therefore statement II is true.

For statement III, first recall what "necessary" and "sufficient" mean.

Saying that A is **necessary** for B means that B cannot hold without A, or equivalently

$$B \Longrightarrow A$$

Saying that A is **sufficient** for B means that A guarantees B, or equivalently

$$A \Longrightarrow B$$

Here, statement III says that for condition (1) to hold, it is necessary but not sufficient that condition (3) does not hold.

Condition (1) is

$$1 < x < 2$$

Condition (3) does not hold when

$$x \leq 2$$

For $x \leq 2$ to be **necessary** for $1 < x < 2$, we need

$$1 < x < 2 \quad \Longrightarrow \quad x \leq 2$$

This is true.

For $x \leq 2$ to be **not sufficient**, it must fail to guarantee $1 < x < 2$. For example, $x = 1$ satisfies

$$x \leq 2$$

but does not satisfy

$$1 < x < 2$$

So condition (3) not holding is necessary for condition (1), but it is not sufficient.

Therefore statement III is true.

Statements II and III are true, while statement I is false, so the correct option is **G**.

Probability & Counting

11. A three-digit code is formed by selecting three different digits from $1, 2, 3, 4, 5, 6$ and placing them in random order.

Each possible code is equally likely.

It is revealed that when the three digits in the code are multiplied together, the result is divisible by 6.

What is the probability that the code contains the digit 6?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{2}{3}$

E $\frac{3}{4}$

F $\frac{5}{6}$

Solution **Answer D**

We are told that the product of the three chosen digits is divisible by 6. Since

$$6 = 2 \times 3$$

the selected digits must supply both a factor of 2 and a factor of 3.

It is convenient to count *sets of three digits* rather than ordered codes. Each set of three distinct digits can be arranged in exactly

$$3! = 6$$

different orders, so every valid selection gives the same number of codes. The ordering therefore does not affect the required probability.

First count the valid selections that contain 6.

Once 6 is chosen, the product is automatically divisible by 6, because the digit 6 itself supplies both a factor of 2 and a factor of 3. We only need to choose the other two digits from $1, 2, 3, 4, 5$, giving

$$\binom{5}{2} = 10$$

valid selections containing 6.

Now count the valid selections that do not contain 6.

The available digits are then

$$1, 2, 3, 4, 5$$

and the product must still contain a factor of 3. The only available multiple of 3 is 3, so 3 must be selected.

We therefore choose the other two digits from

$$1, 2, 4, 5$$

There are

$$\binom{4}{2} = 6$$

ways to do this altogether. However, the pair $\{1, 5\}$ contains no even digit, so the selection $\{1, 3, 5\}$ has no factor of 2 and its product is not divisible by 6.

Thus the number of valid selections not containing 6 is

$$6 - 1 = 5$$

So, among the selections whose product is divisible by 6, there are

$$10 + 5 = 15$$

possible selections in total, of which 10 contain the digit 6.

Hence the required conditional probability is

$$\frac{10}{15} = \frac{2}{3}$$

The correct option is **D**.

Proof & Error Analysis

12. Consider the claim about cubic polynomials:

Claim: If a cubic polynomial f has distinct roots a, b and c, where $a < b < c$, then there exists x with $a < x < b$ such that $f''(x) = 0$

The student attempts to prove the claim and produces the following argument:

I Since $f(a) = f(b) = 0$, there is a number p such that $a < p < b$ and $f'(p) = 0$

II Similarly, there is a number q such that $b < q < c$ and $f'(q) = 0$

III The polynomial f' is quadratic with roots p and q, so $f''\left(\frac{p+q}{2}\right) = 0$

IV Since $a < p < b < q < c$, we have $p - b < 0 < q - b$, and hence

$$\frac{p+q}{2} - b = \frac{(p-b)+(q-b)}{2} < 0$$

V Therefore the claim is true, the claimed x can be chosen as $x = \frac{p+q}{2}$

Which line contains the first error?

A Line **I**

B Line **II**

C Line **III**

D Line **IV**

E Line **V**

F There is no error, the proof is correct as written.

Solution **Answer D**

Lines I and II are valid:

Between the consecutive roots a and b, the graph of the cubic lies entirely on one side of the x-axis. Since it starts on the axis at $x = a$ and returns to the axis at $x = b$, it must turn somewhere between them. Hence there is some p with

$$a < p < b$$

such that

$$f'(p) = 0$$

The same reasoning between the consecutive roots b and c gives some q with

$$b < q < c$$

such that

$$f'(q) = 0$$

So Lines I and II are correct.

Line III is also correct. Since f is cubic, f' is quadratic, and we have just found its two roots p and q.

The turning point of a quadratic lies halfway between its roots, so the turning point of f' has x-coordinate

$$\frac{p+q}{2}$$

At a turning point of f', its gradient is zero. Since the gradient of f' is f'',

$$f''\left(\frac{p+q}{2}\right) = 0$$

The first error occurs in Line IV:

From

$$p < b < q$$

we certainly have

$$p - b < 0 \qquad \text{and} \qquad q - b > 0$$

But one of these numbers is negative and the other is positive, so we cannot determine the sign of their sum.

For example, it could be that

$$p - b = -1 \qquad \text{and} \qquad q - b = 2$$

in which case

$$(p - b) + (q - b) = 1 > 0$$

It could instead be negative, or it could be zero. Therefore

$$\frac{(p - b) + (q - b)}{2} < 0$$

does not follow from the information given.

So Line IV has not shown that

$$\frac{p+q}{2} < b$$

The point where $f''(x) = 0$ is halfway between p and q, but that midpoint could lie on either side of b, or even be equal to b.

Therefore the first error is in Line IV, so the correct option is **D**.

Remark: (Optional)

The claim itself is actually false. For example, take

$$f(x) = x^3 - x$$

Its three distinct roots are

$$-1, \quad 0, \quad 1$$

so here

$$a = -1, \qquad b = 0, \qquad c = 1$$

Differentiating twice gives

$$f'(x) = 3x^2 - 1$$

and

$$f''(x) = 6x$$

Hence

$$f''(x) = 0$$

only when

$$x = 0$$

But $x = 0$ is exactly b, not a value between a and b. Thus there is no x with

$$-1 < x < 0$$

for which $f''(x) = 0$

So the student's argument fails at Line IV, and in this case it is attempting to prove a claim which is itself false.

Integration

13. For each positive integer n, define

$$F(n) = \int_0^1 x^n \, \mathrm{d}x$$

and

$$H(n) = \sum_{r=1}^{n} \frac{1}{F(r)}$$

What is the smallest positive integer n such that $H(n) > 15$?

A 3

B 4

C 5

D 6

E 7

Solution **Answer C**

First evaluate $F(r)$:

$$F(r) = \int_0^1 x^r \, \mathrm{d}x = \left[\frac{x^{r+1}}{r+1}\right]_0^1 = \frac{1}{r+1}$$

Therefore

$$\frac{1}{F(r)} = r + 1$$

and hence

$$H(n) = 2 + 3 + 4 + \cdots + (n+1)$$

This is an arithmetic series with first term $a = 2$, common difference $d = 1$, and n terms.

For an arithmetic series, one form of the sum formula is

$$S_n = \frac{n}{2}\big[2a + (n-1)d\big]$$

Another equivalent form, if the last term is l, is

$$S_n = \frac{n}{2}(a + l)$$

Using the first formula with $a = 2$ and $d = 1$,

$$H(n) = \frac{n}{2}\big[2(2) + (n-1)(1)\big]$$

Therefore

$$H(n) = \frac{n}{2}(n+3) = \frac{n(n+3)}{2}$$

We require

$$H(n) > 15$$

so

$$\frac{n(n+3)}{2} > 15$$

Multiplying by 2,

$$n(n+3) > 30$$

and therefore

$$n^2 + 3n - 30 > 0$$

The corresponding quadratic equation has roots

$$n = \frac{-3 \pm \sqrt{3^2 - 4(1)(-30)}}{2} = \frac{-3 \pm \sqrt{129}}{2}$$

Only the positive root is relevant. Since

$$11^2 < 129 < 12^2$$

we have

$$11 < \sqrt{129} < 12$$

and hence

$$4 < \frac{-3 + \sqrt{129}}{2} < \frac{9}{2}$$

The quadratic is positive for values of n greater than this positive root, so the smallest positive integer that works is

$$n = 5$$

The correct option is **C**.

Alternative Solution:

After finding

$$F(r) = \frac{1}{r+1}$$

we have

$$H(n) = \sum_{r=1}^{n} (r+1)$$

Split this into two standard sums:

$$H(n) = \sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$

Recall that

$$\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$$

and, since the sum contains n copies of 1,

$$\sum_{r=1}^{n} 1 = n$$

Therefore

$$H(n) = \frac{n(n+1)}{2} + n = \frac{n(n+3)}{2}$$

We need

$$\frac{n(n+3)}{2} > 15$$

or equivalently

$$n(n+3) > 30$$

Since the answers are consecutive integers, it is quicker here to test the relevant options than to solve the quadratic.

For $n = 4$,

$$n(n+3) = 4(7) = 28 < 30$$

so $n = 4$ is too small.

For $n = 5$,

$$n(n+3) = 5(8) = 40 > 30$$

so $n = 5$ works.

Also, each additional term in $H(n)$ is positive, so $H(n)$ increases as n increases. Since $n = 4$ fails and $n = 5$ works, the smallest possible value is $n = 5$.

The correct option is **C**.

Logic

14. Let S be the statement:

$$\text{For all real numbers } x, \text{ if } |x| \geq 1 \text{ then } g(x) > 0$$

Which of the following is a correct **negation** of S?

A There exists a real number x with $|x| \geq 1$ such that $g(x) \leq 0$

B There exists a real number x with $|x| > 1$ such that $g(x) < 0$

C For all real numbers x, either $|x| < 1$ or $g(x) > 0$

D For all real numbers x with $|x| \geq 1$, $g(x) \geq 0$

E There exists a real number x such that $|x| < 1$ and $g(x) \leq 0$

F There exists a real number x such that $|x| < 1$ or $g(x) \leq 0$

Solution **Answer A**

There are two standard rules for negating statements of this kind.

First, negating a statement about **all** values changes

$$\text{“for all”} \quad \longleftrightarrow \quad \text{“there exists”}$$

So the negation of

$$\text{For all real numbers } x, \ \ldots$$

must begin

$$\text{There exists a real number } x \text{ such that } \ \ldots$$

We now need to negate what follows.

For a statement of the form

$$\text{if A, then B}$$

the negation is

$$\text{A and not B}$$

This is closely connected to the idea of a counterexample. To show that “if A, then B” is false, we need an example for which A really does hold, but B fails.

Here,

$$\text{A:} \quad |x| \geq 1$$

and

$$\text{B:} \quad g(x) > 0$$

Therefore the negation of the implication is

$$|x| \geq 1 \quad \text{and} \quad \text{not}\big(g(x) > 0\big)$$

The negation of $g(x) > 0$ is

$$g(x) \leq 0$$

The equality case is important: if $g(x) = 0$, then $g(x) > 0$ is already false.

Putting the two negation rules together, the negation of the original statement is

$$\text{There exists a real number } x \text{ such that } |x| \geq 1 \text{ and } g(x) \leq 0$$

In other words, the negation says that there is at least one counterexample to the original claim.

The correct option is **A**.

Proof & Error Analysis

15. A function f is defined for all real x, and $f'(x)$ exists for every real x.

The function f is also **strictly increasing:** whenever $a < b$, also $f(a) < f(b)$.

A student makes the following claim:

Claim: It follows that $f'(x) > 0$ for every real x

The student argues as follows:

Line 1: Suppose, for a contradiction, that $f'(c) \leq 0$ for some real number c

Line 2: If $f'(c) < 0$, then just to the right of c the graph of f falls below $f(c)$, which contradicts f being strictly increasing. Hence $f'(c) = 0$

Line 3: So the tangent to the graph of f at $x = c$ is horizontal

Line 4: A horizontal tangent at $x = c$ means there are numbers $a < c < b$ with $f(a) \geq f(c)$ or with $f(b) \leq f(c)$

Line 5: Either possibility contradicts f being strictly increasing. Hence $f'(x) > 0$ for every real x

Which of the following best describes the argument?

A The argument is correct, and the claim is true

B The first error occurs in Line 2

C The first error occurs in Line 3

D The first error occurs in Line 4

E The first error occurs in Line 5

F The argument contains an error, but the claim is nevertheless true

Solution **Answer D**

Line 1 is a valid way to begin a proof by contradiction. The student assumes that

$$f'(c) \leq 0$$

for some real number c

Line 2 is also correct. If

$$f'(c) < 0$$

then the graph is sloping downwards at $x = c$. Moving a sufficiently small distance to the right of c would therefore make the value of f smaller than $f(c)$. This is impossible for a strictly increasing function.

Hence, under the assumption in Line 1, the only remaining possibility is

$$f'(c) = 0$$

Line 3 is therefore correct. If

$$f'(c) = 0$$

then the tangent to the graph at $x = c$ has gradient 0, so it is horizontal.

The first error is in Line 4.

A horizontal tangent does not imply either of the inequalities claimed there. Consider

$$f(x) = x^3$$

and take

$$c = 0$$

Then

$$f'(x) = 3x^2$$

so

$$f'(0) = 0$$

Thus the tangent at $x = 0$ is horizontal.

However, for any $a < 0 < b$,

$$f(a) = a^3 < 0 = f(0) < b^3 = f(b)$$

Therefore

$$f(a) < f(c)$$

and

$$f(b) > f(c)$$

So neither

$$f(a) \geq f(c)$$

nor

$$f(b) \leq f(c)$$

holds.

This directly contradicts the assertion made in Line 4, so Line 4 is the first error.

The same example also shows that the original claim is false: $f(x) = x^3$ is strictly increasing, but

$$f'(0) = 0$$

Therefore the first error occurs in Line 4, so the correct option is **D**.

Sequences & Series

16. Let n be an integer with $n \geq 6$.

In the expansion of $(1+x)^n$ the coefficients of x^4, x^5 and x^6, in that order, are consecutive terms of an arithmetic sequence.

What is the sum of all possible values of n?

A 13

B 14

C 20

D 21

E 22

F 28

Solution **Answer D**

By the binomial theorem,

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{r}x^r + \cdots$$

Therefore the coefficients of x^4, x^5 and x^6 are respectively

$$\binom{n}{4}, \qquad \binom{n}{5}, \qquad \binom{n}{6}$$

Note that the notation $\binom{n}{r}$ is also sometimes written ${}^nC_r$.

Since these three coefficients are consecutive terms of an arithmetic sequence, their successive differences must be equal:

$$\binom{n}{5} - \binom{n}{4} = \binom{n}{6} - \binom{n}{5}$$

Rearranging,

$$2\binom{n}{5} = \binom{n}{4} + \binom{n}{6}$$

Using

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

gives

$$2\left(\frac{n!}{5!(n-5)!}\right) = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$$

Since $n! \neq 0$, we can divide every term by $n!$:

$$\frac{2}{5!(n-5)!} = \frac{1}{4!(n-4)!} + \frac{1}{6!(n-6)!}$$

Now multiply throughout by $(n-4)!$:

$$\frac{2(n-4)!}{5!(n-5)!} = \frac{(n-4)!}{4!(n-4)!} + \frac{(n-4)!}{6!(n-6)!}$$

The middle term simplifies immediately:

$$\frac{(n-4)!}{4!(n-4)!} = \frac{1}{4!}$$

For the first term,

$$(n-4)! = (n-4)(n-5)!$$

so

$$\frac{(n-4)!}{(n-5)!} = n-4$$

For the final term,

$$(n-4)! = (n-4)(n-5)(n-6)!$$

so

$$\frac{(n-4)!}{(n-6)!} = (n-4)(n-5)$$

Hence

$$\frac{2(n-4)}{5!} = \frac{1}{4!} + \frac{(n-4)(n-5)}{6!}$$

Multiplying throughout by 6! gives

$$\frac{2 \cdot 6!}{5!}(n-4) = \frac{6!}{4!} + (n-4)(n-5)$$

Now

$$\frac{6!}{5!} = 6$$

and

$$\frac{6!}{4!} = 6 \cdot 5 = 30$$

so

$$12(n-4) = 30 + (n-4)(n-5)$$

Expanding,

$$12n - 48 = 30 + n^2 - 9n + 20$$

Therefore

$$n^2 - 21n + 98 = 0$$

Factorising,

$$(n-7)(n-14) = 0$$

Thus

$$n = 7 \qquad \text{or} \qquad n = 14$$

Both values satisfy $n \geq 6$, so the sum of all possible values of n is

$$7 + 14 = 21$$

The correct option is **D**.

Once the quadratic

$$n^2 - 21n + 98 = 0$$

has been obtained, the sum of its roots can be read off without factorising, although this answers the question only because both roots, 7 and 14, are possible values of n. For a quadratic $an^2 + bn + c = 0$, the sum of the roots is

$$-\frac{b}{a}$$

so here the sum of the two possible values of n is immediately

$$-\frac{-21}{1} = 21$$

Logic

17. Let a be a real number. Consider the following statement:

$$\text{For all real numbers } x, \text{ there exists a real number } y \text{ such that}$$
$$a \leq y \leq a + \pi \text{ and } \sin y = \sin x$$

Which condition on a is **necessary and sufficient** for the statement to be true?

A $\sin a = 0$

B $\cos a = 0$

C $\sin 2a = 0$

D $\cos 2a = 0$

E $\sin a = \cos a$

F $\sin a = -\cos a$

Solution **Answer B**

As x runs through all real numbers, $\sin x$ takes every value from -1 to 1.

So the statement is asking for the interval

$$a \leq y \leq a + \pi$$

to contain enough of the sine graph for $\sin y$ to take every possible value between -1 and 1.

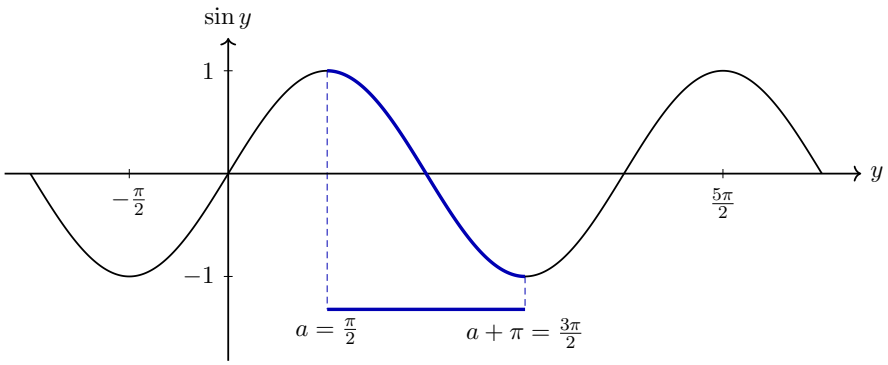
In particular, the interval must contain some point where

$$\sin y = 1$$

and also some point where

$$\sin y = -1$$

These are respectively a maximum and a minimum of the sine graph.



Successive maxima and minima of the sine graph are π apart. For example,

$$\sin\left(\frac{\pi}{2}\right) = 1$$

and

$$\sin\left(\frac{3\pi}{2}\right) = -1$$

More generally, a maximum and the neighbouring minimum are always separated by exactly π.

But the interval from a to $a+\pi$ also has length exactly π. Therefore, if it is to contain both a maximum and a minimum, these two points must occur at the two endpoints of the interval.

Hence

$$\sin a = 1$$

and

$$\sin(a+\pi) = -1$$

or else

$$\sin a = -1$$

and

$$\sin(a+\pi) = 1$$

In either case,

$$\sin a = \pm 1$$

Using

$$\sin^2 a + \cos^2 a = 1$$

we obtain

$$\cos^2 a = 1 - \sin^2 a = 0$$

so

$$\cos a = 0$$

Thus $\cos a = 0$ is a necessary condition.

We must also check that it is sufficient.

Suppose

$$\cos a = 0$$

Then

$$\sin a = \pm 1$$

So a is a maximum or a minimum of the sine graph. Since increasing the input by π changes the sign of the sine,

$$\sin(a+\pi) = -\sin a$$

Therefore $a+\pi$ is the neighbouring minimum or maximum.

As y moves from a to $a+\pi$, the sine graph moves continuously from 1 to -1, or from -1 to 1. Hence it takes every value between -1 and 1.

Since $\sin x$ must always lie between -1 and 1, for every real x there is therefore some y with

$$a \leq y \leq a+\pi$$

such that

$$\sin y = \sin x$$

So $\cos a = 0$ is both necessary and sufficient.

The correct option is **B**.

Logic

18. Let p and q be real constants, and let

$$f(x) = x^3 + px^2 + qx$$

Consider the statement

S: There exists a real number x such that $f'(x) < 0$

Which of the following conditions is **necessary** but **not sufficient** for S?

A $q < 0$

B $q \leq 0$

C $3q < p^2$

D $3q \leq p^2$

E $p^2 > 0$

F $p^2 \geq 3|q|$

Solution **Answer D**

First translate the logic in the question.

A condition is **necessary** for S if S cannot be true without that condition also being true.

A condition is **not sufficient** for S if the condition can be true while S is false.

So we are looking for a condition which must hold whenever there is some real x with

$$f'(x) < 0$$

but which does not, by itself, guarantee that such an x exists.

Differentiate:

$$f'(x) = 3x^2 + 2px + q$$

Since $f'(x)$ is a quadratic with positive x^2-coefficient, it has a minimum. Completing the square makes that minimum easy to identify:

$$\begin{aligned} f'(x) &= 3x^2 + 2px + q \\ &= 3\left(x^2 + \frac{2p}{3}x\right) + q \\ &= 3\left(x + \frac{p}{3}\right)^2 - \frac{p^2}{3} + q \end{aligned}$$

Thus

$$f'(x) = 3\left(x + \frac{p}{3}\right)^2 + q - \frac{p^2}{3}$$

The squared term is always non-negative, and it is 0 when

$$x = -\frac{p}{3}$$

Therefore the minimum possible value of $f'(x)$ is

$$q - \frac{p^2}{3}$$

For there to exist some real x for which $f'(x) < 0$, this minimum must be strictly negative:

$$q - \frac{p^2}{3} < 0$$

Multiplying by 3 gives

$$3q < p^2$$

So S is true exactly when

$$3q < p^2$$

This means option C is both necessary *and* sufficient for S, so it is not the answer being asked for.

Now consider option D:

$$3q \leq p^2$$

Whenever S is true, we have

$$3q < p^2$$

which certainly implies

$$3q \leq p^2$$

So option D is necessary.

However, it is not sufficient, because equality is also allowed. For example, take

$$p = 0, \qquad q = 0$$

Then

$$3q \leq p^2$$

is true, since both sides are 0. But

$$f'(x) = 3x^2$$

and therefore

$$f'(x) \geq 0$$

for every real x

So there is no real x for which $f'(x) < 0$, and S is false.

Thus $3q \leq p^2$ is necessary but not sufficient for S.

The correct option is **D**.

Algebra

19. A real number x satisfies

$$x^2 - 3x + 1 = 0$$

What is the value of $x^6 + \dfrac{1}{x^6}$?

A 288

B 306

C 320

D 322

E 324

Solution **Answer D**

The expression

$$x^6 + \frac{1}{x^6}$$

contains matching positive and negative powers of x. This suggests trying to find

$$x + \frac{1}{x}$$

rather than solving explicitly for x.

We are given

$$x^2 - 3x + 1 = 0$$

Since $x = 0$ does not satisfy this equation, we may divide throughout by x:

$$x - 3 + \frac{1}{x} = 0$$

Hence

$$x + \frac{1}{x} = 3$$

Now square both sides:

$$\left(x + \frac{1}{x}\right)^2 = 3^2$$

Expanding,

$$x^2 + 2 + \frac{1}{x^2} = 9$$

Therefore

$$x^2 + \frac{1}{x^2} = 7$$

Square again:

$$\left(x^2 + \frac{1}{x^2}\right)^2 = 7^2$$

Expanding,

$$x^4 + 2 + \frac{1}{x^4} = 49$$

Hence

$$x^4 + \frac{1}{x^4} = 47$$

We now have expressions involving the second and fourth powers. Multiplying them produces the sixth powers:

$$\left(x^2 + \frac{1}{x^2}\right)\left(x^4 + \frac{1}{x^4}\right)$$

Expanding,

$$x^6 + x^2 + \frac{1}{x^2} + \frac{1}{x^6}$$

Using

$$x^2 + \frac{1}{x^2} = 7$$

this becomes

$$x^6 + \frac{1}{x^6} + 7$$

But the product is also

$$7 \times 47 = 329$$

Therefore

$$x^6 + \frac{1}{x^6} + 7 = 329$$

so

$$x^6 + \frac{1}{x^6} = 322$$

The correct option is **D**.

Remark: (Exam technique)

Questions of this form are often traps in disguise. It is possible to solve the quadratic first and then substitute

$$x = \frac{3 \pm \sqrt{5}}{2}$$

but raising either of these expressions to the sixth power would create a large amount of unnecessary algebra.

The paired form

$$x^n + \frac{1}{x^n}$$

is the important clue. When the original equation has a non-zero constant term, dividing by a suitable power of x can often reveal an expression such as

$$x + \frac{1}{x}$$

which can then be used to build the required higher powers.

In a timed test, questions like this can consume a disproportionate amount of time if the intended simplification is not spotted. Before committing to a long calculation, it is worth checking whether the form of the required expression suggests a shorter route.

Polynomials

20. The equation

$$x^3 - 3x + p = 0$$

where p is a real constant, has three distinct real solutions for x.

Find the complete set of values of p

A $-3 < p < 3$

B $-2 < p < 2$

C $-2 \leq p \leq 2$

D $p < -2$ or $p > 2$

E $0 < p < 4$

F $-4 < p < 0$

Solution **Answer B**

Let

$$f(x) = x^3 - 3x + p$$

The real solutions of

$$x^3 - 3x + p = 0$$

are exactly the x-coordinates where the graph of $y = f(x)$ crosses or touches the x-axis.

To have three distinct real solutions, the cubic must therefore meet the x-axis at three different points.

First find the stationary points:

$$f'(x) = 3x^2 - 3$$

Setting $f'(x) = 0$,

$$3x^2 - 3 = 0$$

so

$$x^2 = 1$$

and hence

$$x = -1 \qquad \text{or} \qquad x = 1$$

The cubic rises to a local maximum at $x = -1$, then falls to a local minimum at $x = 1$.

Their heights are

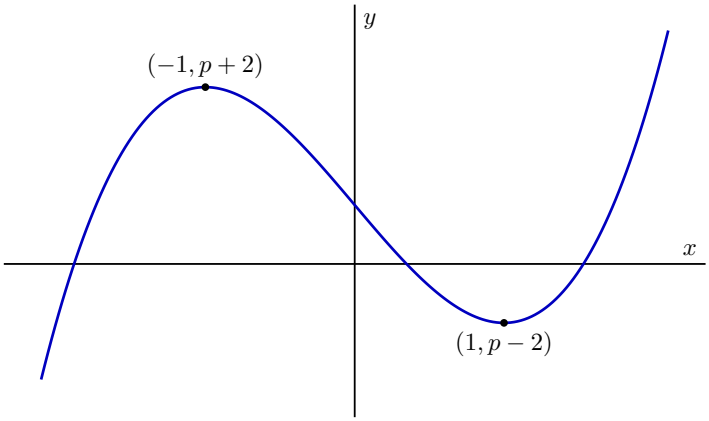
$$f(-1) = (-1)^3 - 3(-1) + p = p + 2$$

and

$$f(1) = 1^3 - 3(1) + p = p - 2$$

So the stationary points are

$$(-1, p + 2) \qquad \text{and} \qquad (1, p - 2)$$



Since the coefficient of x^3 is positive, the graph eventually goes downwards without bound to the left and upwards without bound to the right.

For the graph to cross the x-axis three times, its local maximum must lie above the x-axis, while its local minimum must lie below the x-axis.

Therefore we require

$$p + 2 > 0$$

and

$$p - 2 < 0$$

The first inequality gives

$$p > -2$$

The second gives

$$p < 2$$

Combining these,

$$-2 < p < 2$$

The inequalities must be strict. For example, if $p = 2$, then the local minimum lies exactly on the x-axis:

$$f(1) = 0$$

so the graph touches the axis at $x = 1$ rather than crossing it there. This produces a repeated root, not three distinct real roots.

Similarly, $p = -2$ makes the local maximum lie on the x-axis and again gives a repeated root.

Hence the complete set of values is

$$-2 < p < 2$$

The correct option is **B**.